

TDT4230 Graphics and Visualization

Procedural Terrain Generation

Simone D'Assisi
2025/2026



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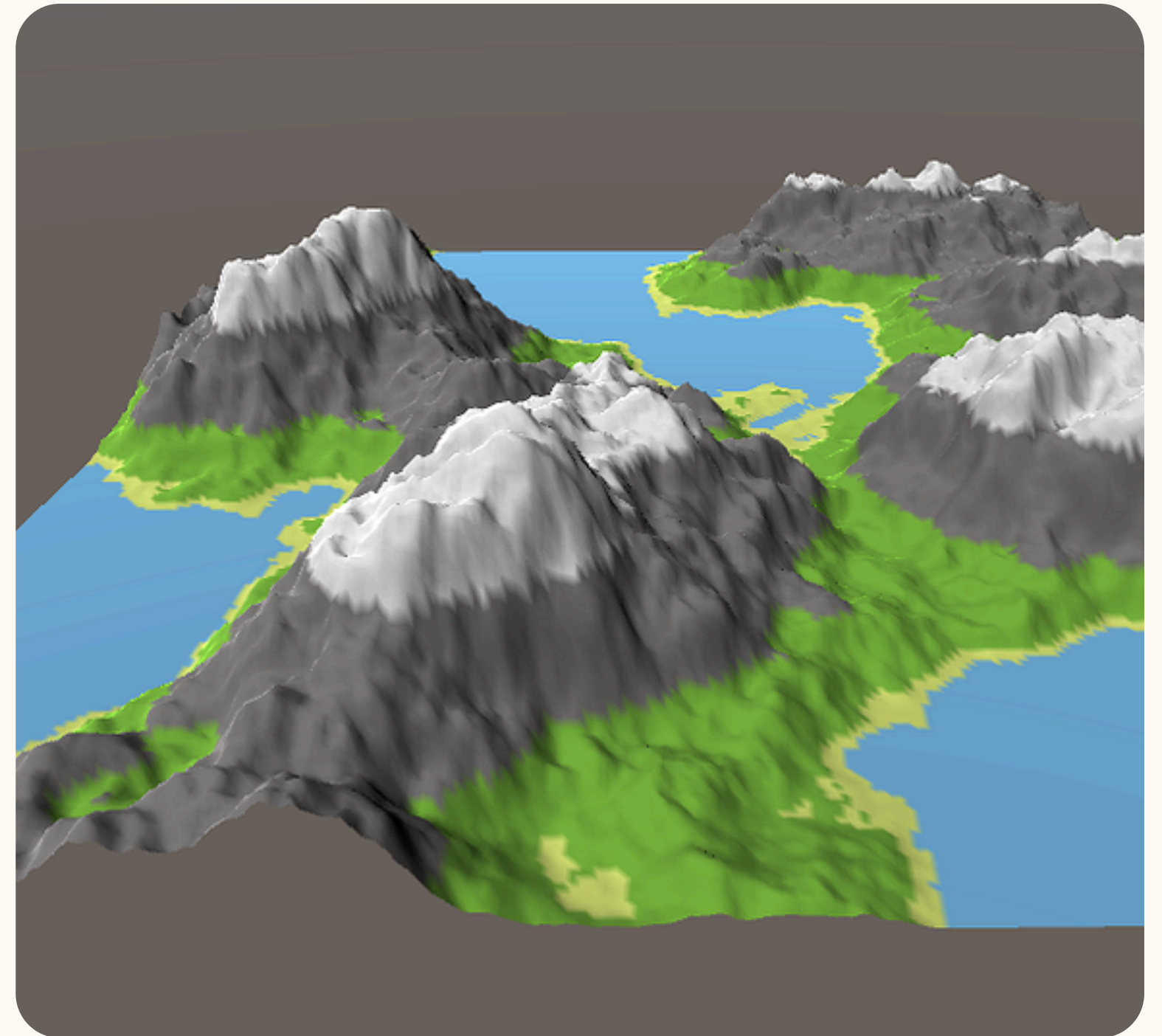


| PROJECT OBJECTIVES

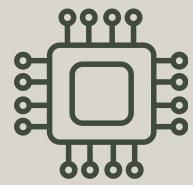
Beyond Static Assets

Development of a 3D world generated entirely via code, overcoming the limitations of pre-built models and ensuring infinite variability.

- 🌿 **Morphological Realism:** Simulation of mountain ranges and biomes.
- 🌿 **Optimization:** Managing thousands of objects via GPU.
- 🌿 **Immersion:** Dynamic day/night cycle and weather.



| ARCHITECTURE



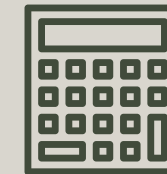
OpenGL 3.3

Core Profile for a shader-based pipeline, utilizing VAOs, VBOs, and EBOs.



C++ & GLSL

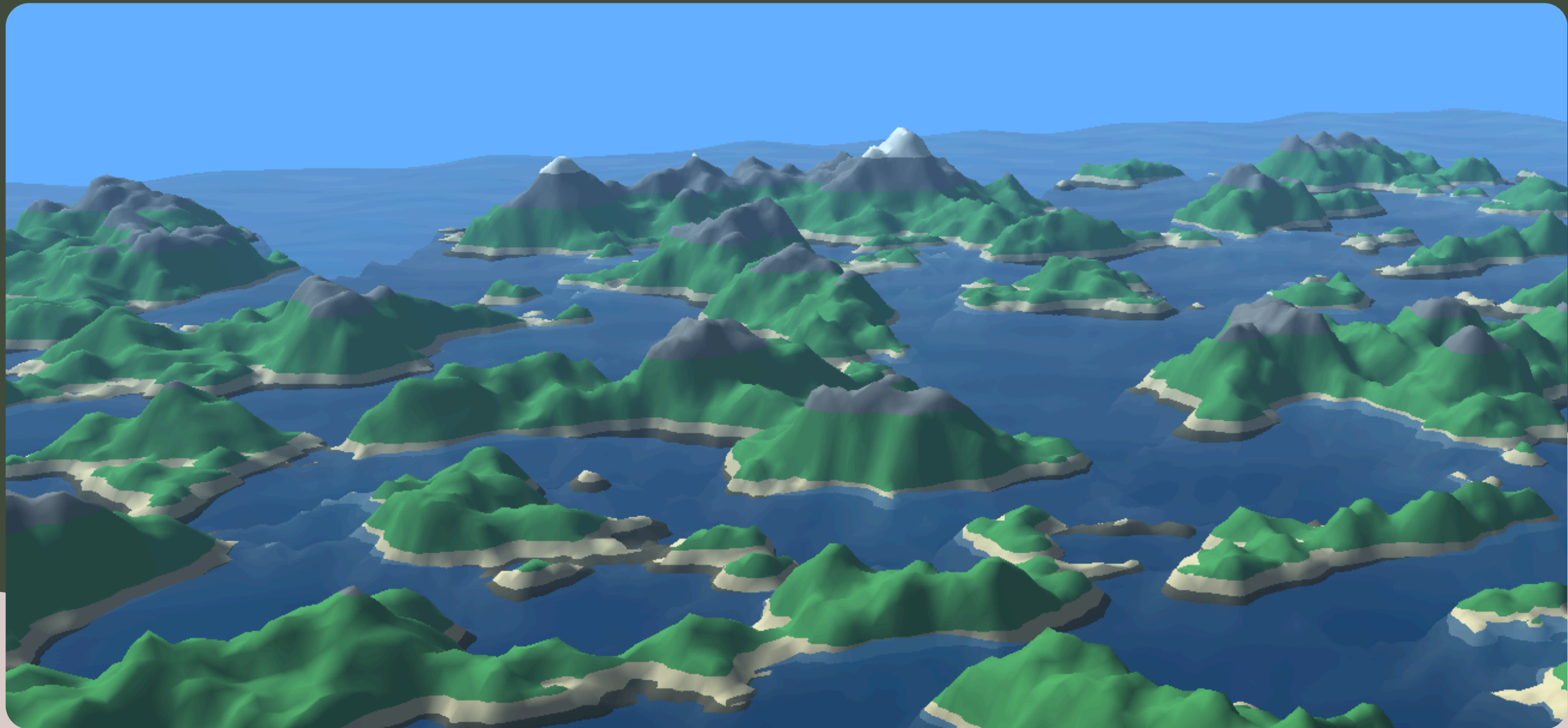
System logic in C++ and heavy mathematical computations (noise, lighting) delegated to the GPU.



GLM & TinyOBJ

Handling of linear algebra (MVP matrices) and parsing of external 3D models.

Procedural Content Generation



| NOISE AND SPATIAL COHERENCE

Fractal Brownian Motion (FBM)

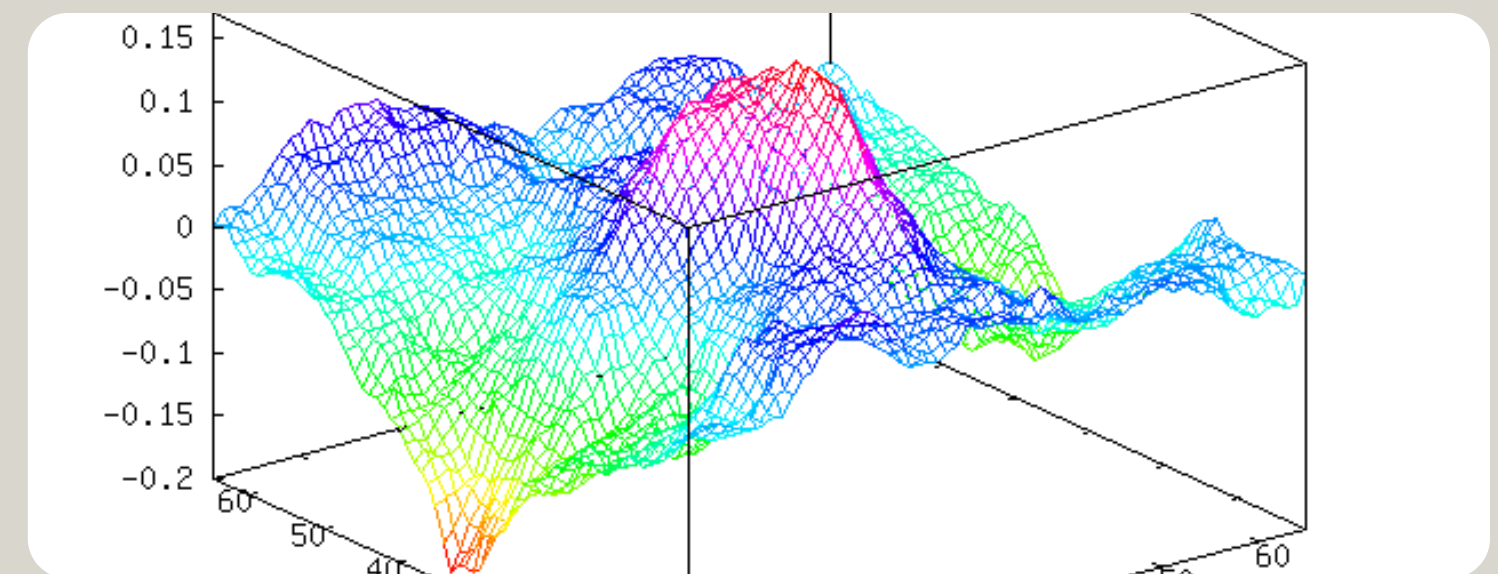
To achieve the jagged appearance of mountains, we layer Perlin Noise (octaves) with increasing frequency and decreasing amplitude.

$$H(x, z) = \sum_{i=0}^{n-1} \text{amplitude}_i \times \text{noise}(\text{frequency}_i \times (x, z))$$

Implementation: 6 octaves for an optimal balance between detail and performance.

Mesh Construction

A grid of vertices where the Y coordinate is extracted from the FBM function. Use of Element Buffer Objects (EBO) to share vertices between adjacent faces and save memory.

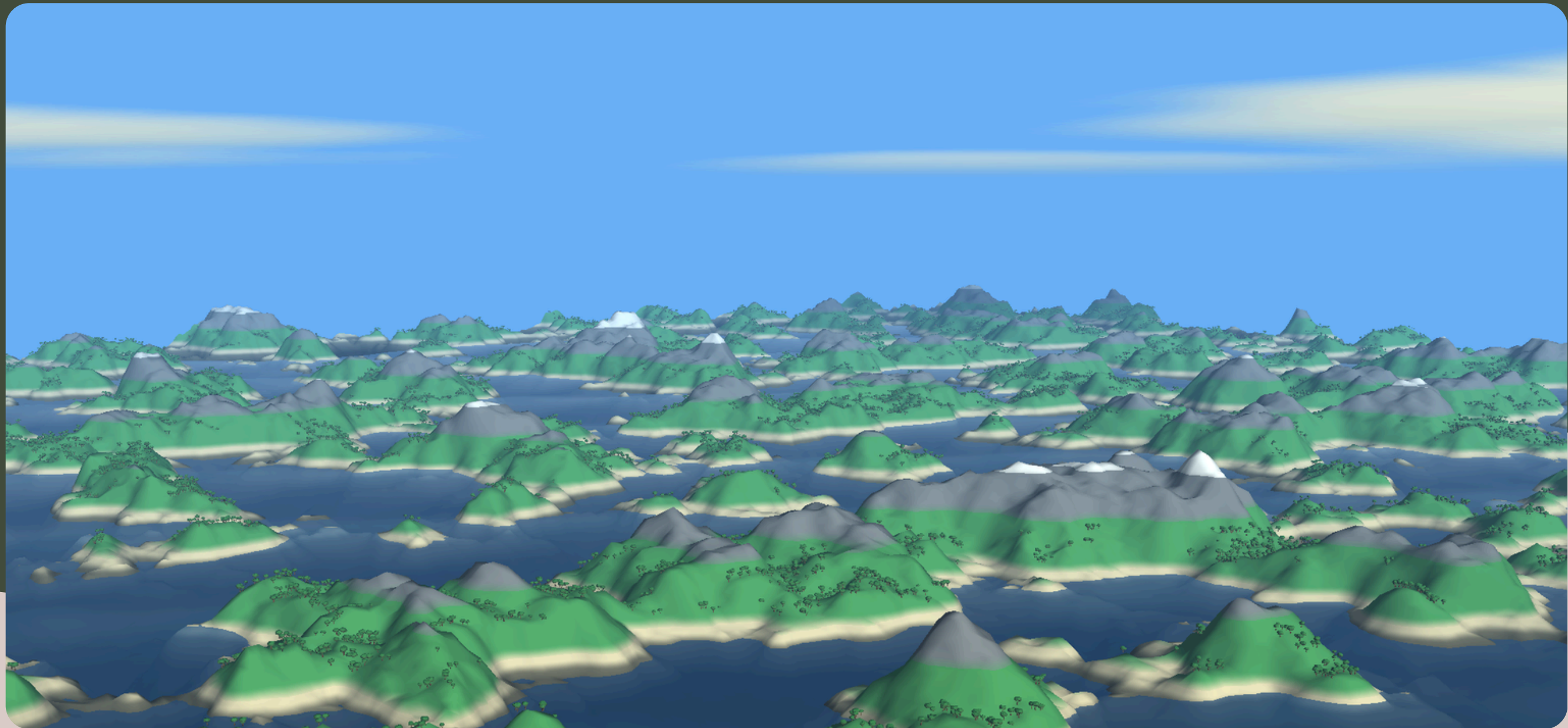


| BIOME CLASSIFICATION

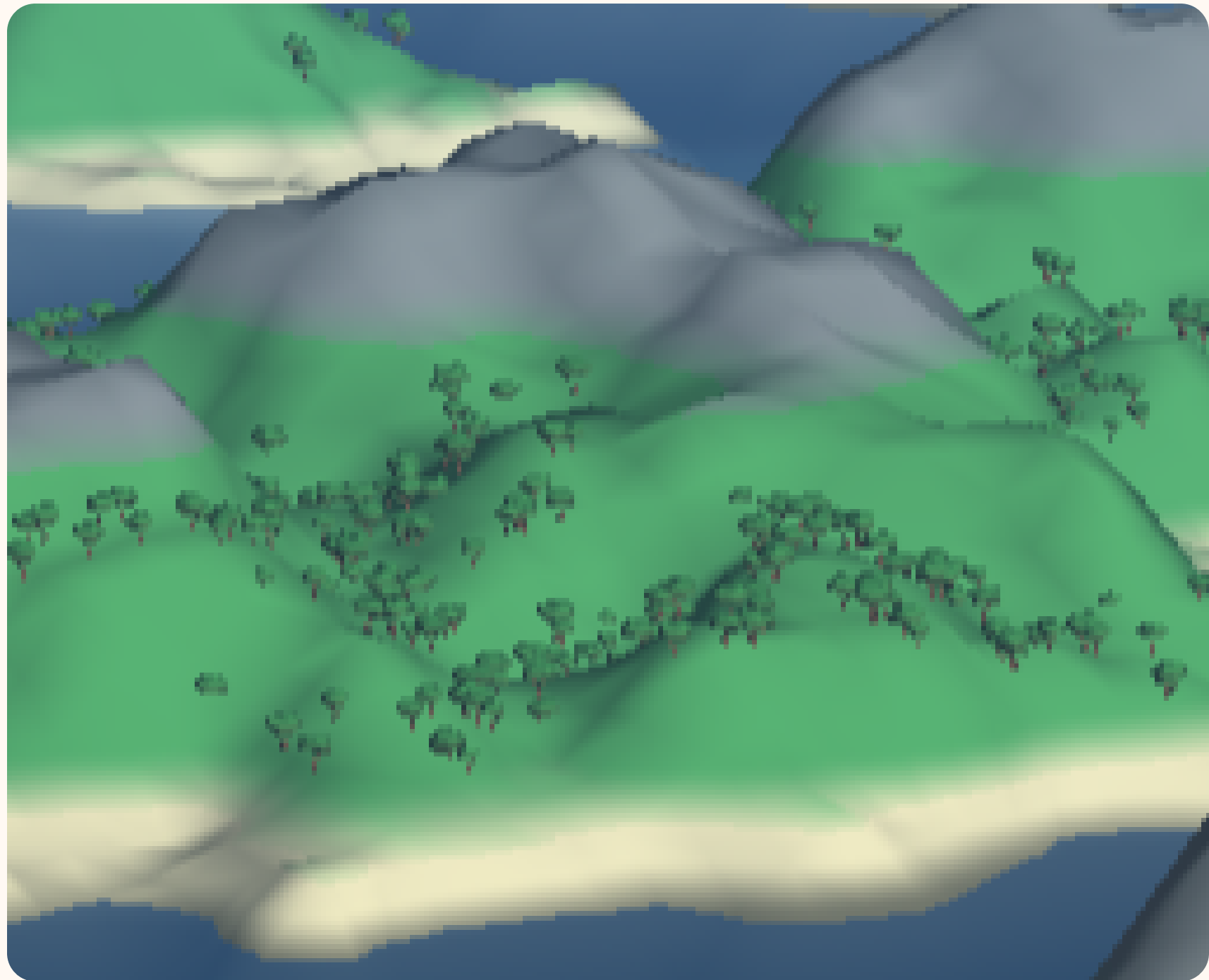
Instead of static textures, the Fragment Shader assigns dynamic colors based exclusively on the altitude (Y coordinate) of the fragment.

Elevation Range	Biome	Visual Representation
$Y < 4.0$	Deep/Shallow Sea	From dark sandy to cyan blue
$4.0 \leq Y < 6.0$	Beach	Light yellow sand
$6.0 \leq Y < 15.0$	Grassland	Green vegetation
$15.0 \leq Y < 25.0$	Rock/Mountain	Dark grey stone
$Y \geq 25.0$	Snow Peaks	Pure white snow

Vegetation and Optimization



| HARDWARE INSTANCING



Massive Drawing in a Single Call

Rendering thousands of individual trees would cause a CPU-GPU bottleneck. Instancing solves this problem:

- 🌿 **Dedicated VBO:** Contains transformation matrices for each instance.
- 🌿 **glDrawElementsInstanced:** A single instruction to draw the entire forest.
- 🌿 **Immersion:** Trees only grow on flat surfaces (analysis of the normal vector N_y).

| PHONG LIGHTING MODEL

Diffuse Reflection

Simulates the impact of light on rough surfaces (Lambertian).

$$I_d = \max \left(\vec{N} \times \vec{L}, 0 \right) \times C_{light}$$

Specular Reflection

Creates intense highlights, especially on water. High shininess exponent ($\alpha=128.0$).

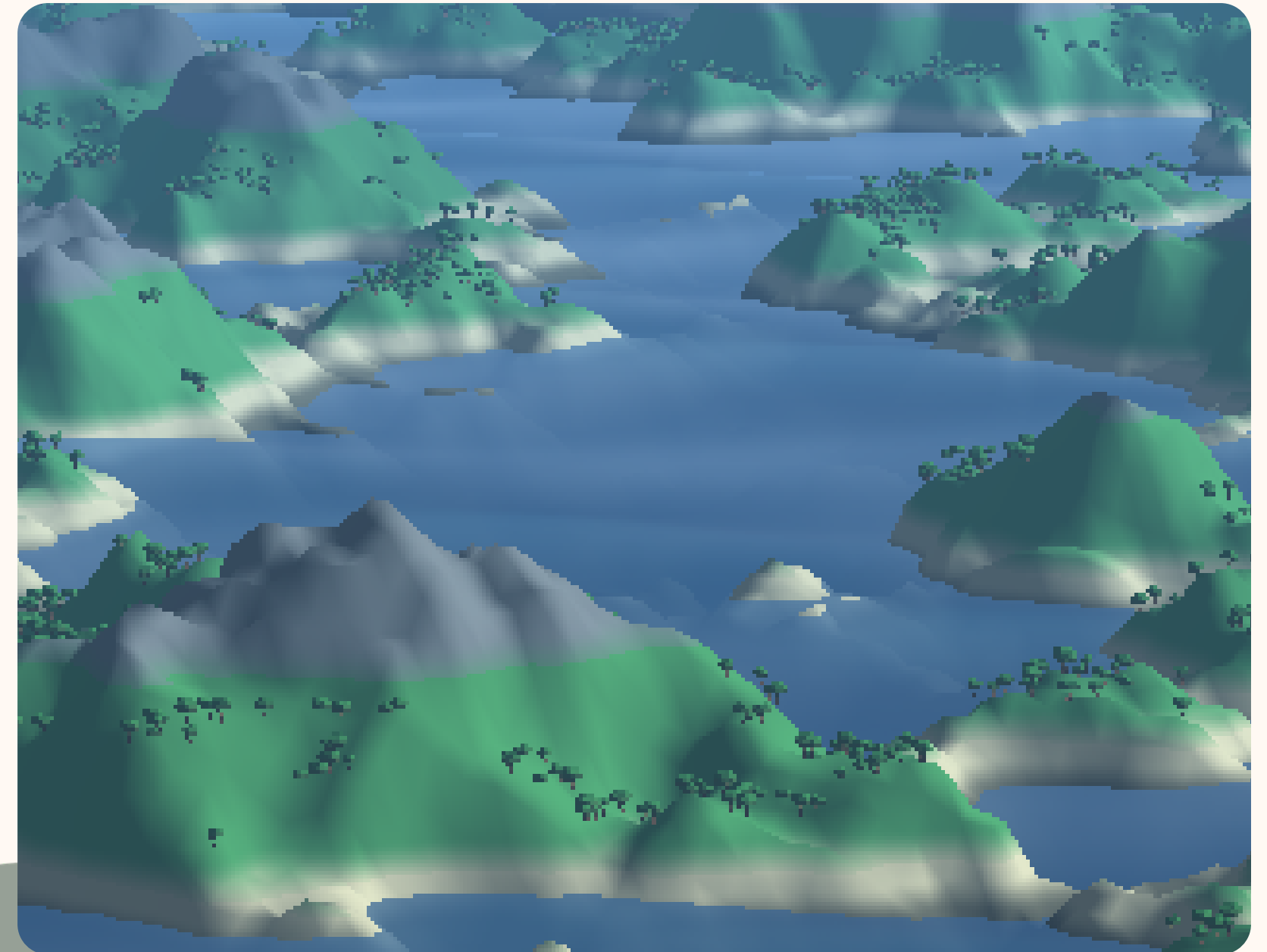
$$I_s = \left(\max \left(\vec{V} \times \vec{R}, 0 \right) \right)^\alpha \times C_{light}$$

| DYNAMIC WATER SIMULATION

Gerstner Waves

Displacement of vertices along X, Y, and Z to create realistic wave profiles with sharp crests.

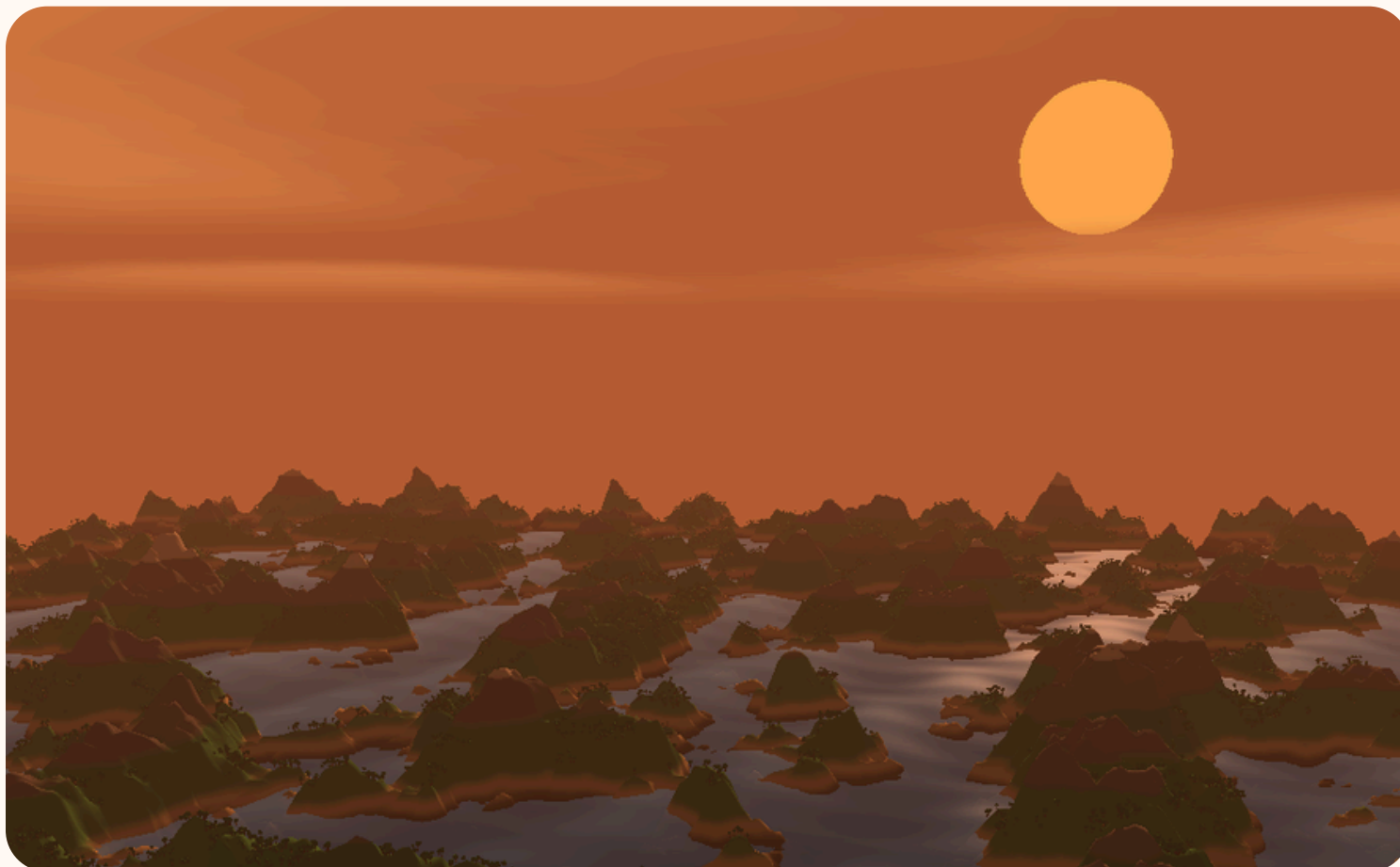
- **Cyclic Displacement:** Vertices move in a circular motion.
- **Analytical Normals:** Calculated in the Vertex Shader using partial derivatives for perfect lighting.
- **Procedural Foam:** Generated dynamically only on the highest crests.



| ATMOSPHERE AND DAY/NIGHT CYCLE

Solar Trajectory

A directional light rotates around the scene, modifying global colors based on the angle of incidence.



Daylight: Intense sky blue.

Dawn/Dusk: Warm, orange tones.

Night: Deep blue and soft ambient light.



| NOISE AND SPATIAL COHERENCE

Volumetric Clouds

System of 15 overlapping horizontal quads. Each layer samples an FBM function with vertical offsets, creating the illusion of depth.



Exponential Fog

Models the physical scattering of light over vast distances, hiding the edges of the procedural mesh.

$$FogFactor = clamp(e^{-(distance \times density)}, 0.0, 1.0)$$

| SHADER ARCHITECTURE

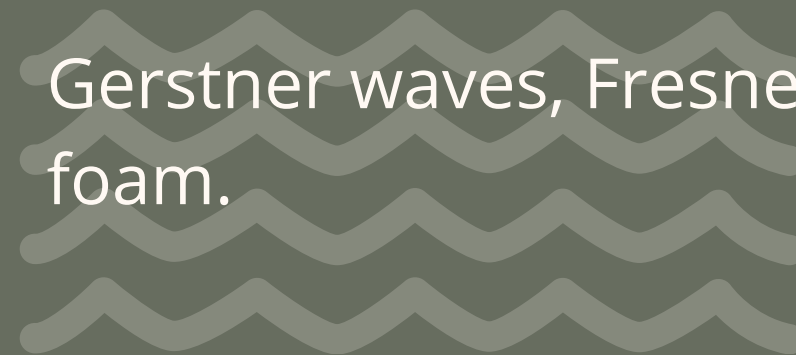
Terrain Shader

Phong lighting, Y-based biomes, and fog.



Water Shader

Gerstner waves, Fresnel effect, and foam.



Tree Shader

Massive instancing and world-space coordinates.



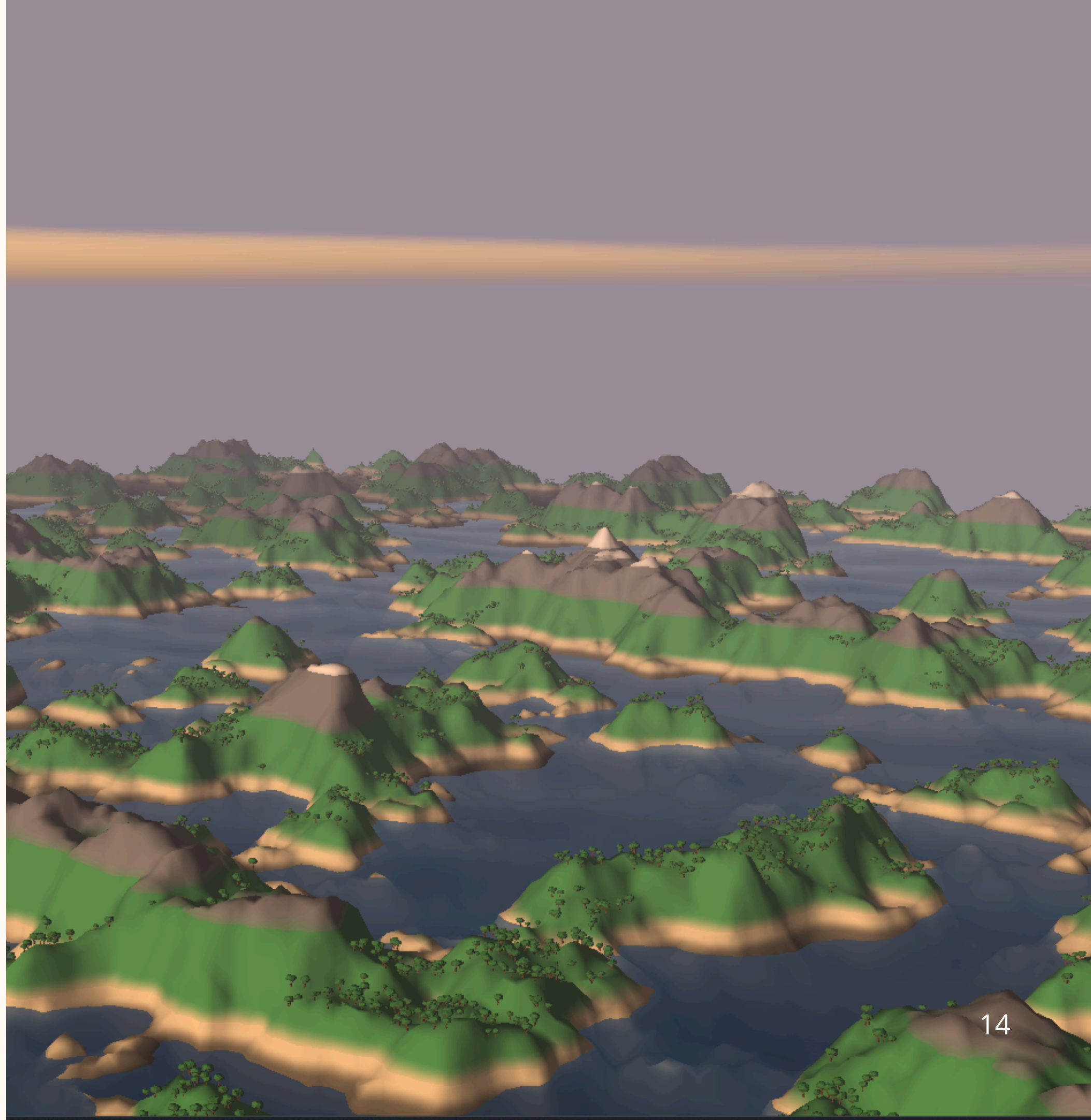
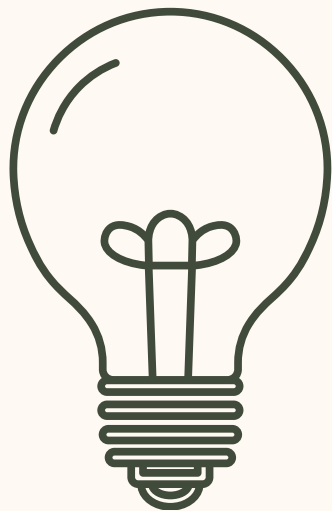
Cloud Shader

Alpha blending and animated fractal noise.



| CONCLUSIONS

The project demonstrates the effectiveness of procedural techniques for creating complex worlds in real-time with minimal CPU overhead.



Thank you for your attention!



[Link to the repository](#)



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